

The Citation: A Research Document on Its History, Failure Modes, and the Frontier of Graded Truth

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1. What a Citation Actually Is

A citation is commonly understood as a pointer — a reference that connects one piece of writing to another. In practice it is considerably more than that. A citation is an assertion. When an author writes "protein X inhibits pathway Y (Smith et al., 2019)," they are not merely acknowledging Smith et al. They are claiming that Smith et al. provides evidence sufficient to support the assertion being made. The citation is doing epistemic work: it is transferring the authority of the cited source to the current claim.

This distinction — between citation as pointer and citation as assertion — is the foundation of everything that follows. A pointer can be verified by checking whether the source exists. An assertion requires reading the source and determining whether it actually supports what is being claimed. The entire citation integrity problem, across science, law, and AI, flows from the gap between these two things.

2. A Brief History of the Citation

The practice of citing sources is older than the modern academic paper. Ancient scholars referenced prior authorities — Aristotle cited Plato, Galen cited Hippocrates — but these were discursive acknowledgments embedded in prose, not formal bibliographic apparatus. The modern citation system, with its standardised author-date or numbered reference formats, emerged gradually between the seventeenth and twentieth centuries alongside the institutionalisation of scientific publishing. ¹

The first scientific journals — the *Philosophical Transactions of the Royal Society* (1665) and the *Journal des Sçavans* (1665) — began the practice of situating new findings within prior work. But it was not until the late nineteenth and early twentieth centuries that citation became a formalised scholarly obligation rather than a courtesy. The American Psychological Association published its first style guide in 1929. The Modern Language Association followed in 1951. The Chicago Manual of Style, which had existed since 1906, codified footnote and bibliography conventions that became standard in the humanities. ¹

The decisive shift in how citations were understood as data — rather than merely as acknowledgments — came with Eugene Garfield's creation of the Science Citation Index in

1960. Garfield's insight was that citations could be measured, mapped, and used to assess the influence and reliability of scientific work. The number of times a paper was cited became a proxy for its importance. This gave rise to bibliometrics, scientometrics, and eventually the impact factor — a metric that now governs academic careers, journal prestige, and funding allocation worldwide. ¹

The unintended consequence of Garfield's innovation was that citations became currency. Once citation counts could be measured and rewarded, the incentives around citation behaviour changed. Authors began citing strategically — to acknowledge allies, to avoid citing critics, to inflate the apparent consensus around their own work. The citation, which had begun as a scholarly courtesy and evolved into a scholarly obligation, became a social instrument.

3. The Documented Failure Modes of Citation

The scientific literature on citation failure is itself substantial and well-documented. The failures cluster into five categories, each with distinct mechanisms and consequences.

3.1 Fabrication

The most basic failure is the citation that does not exist. Before large language models, fabricated citations were relatively rare and typically the result of deliberate fraud or careless error — a misremembered author name, a wrong year, a paper that was never published. The advent of LLMs has changed this dramatically. A 2026 study published on arXiv estimated **146,932 hallucinated citations in academic papers published in 2025 alone**, based on analysis of four major research repositories. ² An analysis of NeurIPS 2025 papers found over 100 AI-hallucinated citations in 51 accepted papers at one of the world's most prestigious AI conferences. ³

The scale of fabrication is now large enough to constitute a systemic threat to the integrity of the scientific record. The papers that contain these citations are real, peer-reviewed, and indexed. The citations they contain point to sources that do not exist. Every paper that subsequently cites those papers inherits the contamination.

3.2 Misrepresentation

More prevalent and harder to detect than fabrication is misrepresentation — the citation of a real paper in a way that does not accurately reflect what the paper found. A 2018 study in the *Proceedings of the National Academy of Sciences* documented the range of "spin" practices in biomedical literature, from selective reporting of outcomes to misleading interpretation of statistically non-significant results. ⁴ The cited paper exists; the claim being supported by it does not follow from what the paper actually demonstrated.

Misrepresentation takes several forms. An author may cite a paper that found a weak or qualified association as if it demonstrated a strong causal effect. They may cite a study conducted in a specific population (elderly men with metabolic syndrome) as if its findings apply to the general population. They may cite a paper's abstract without reading the limitations section, which may substantially qualify the headline finding. They may cite a preliminary study as if it were replicated consensus.

The Office of Research Integrity at the US Department of Health and Human Services has documented multiple forms of ethically questionable citation practice, including citing papers regardless of relevance to inflate self-citation counts, and citing papers that contradict the author's argument while framing them as supporting it. ⁵

3.3 Amplification and Distortion

The most consequential failure mode at the level of the scientific literature as a whole is what Steven Greenberg, in a landmark 2009 study in the *BMJ*, called **citation distortion through amplification**. Greenberg constructed the complete citation network for a specific scientific belief — that β -amyloid protein is produced by and injures skeletal muscle in inclusion body myositis — and found 242 papers and 675 citations addressing this belief, generating 220,553 citation paths. ⁶

What Greenberg found was that the belief had achieved the status of established fact through three mechanisms operating in the citation network:

Citation bias — papers that refuted or weakened the belief were systematically under-cited relative to papers that supported it, regardless of their methodological quality.

Amplification — the belief was cited and re-cited by papers that presented no new data addressing it, expanding the apparent evidentiary base without adding any actual evidence.

Invention — through repeated citation, a hypothesis was converted into an established fact. Papers that originally stated "X may be involved in Y" were cited by later papers as if they had demonstrated "X causes Y," and those later papers were in turn cited as if the causal claim were settled.

The result was a scientific consensus built not on the weight of evidence but on the topology of the citation network. The belief had 220,553 citation paths supporting it and was accepted as fact by the field — despite the primary data being weak and contested.

"Citation is both an impartial scholarly method and a powerful form of social communication. Through distortions in its social use that include bias, amplification, and invention, citation can be used to generate information cascades resulting in unfounded authority of claims." — Greenberg, *BMJ* 2009 ⁶

3.4 Error Propagation Through Retraction

A related failure mode is the propagation of retracted findings through citation chains. A 2016 study in *Research Integrity and Peer Review* constructed the complete citation network for a paper published in *Nature* and subsequently retracted, finding that retracted results continue to be cited and propagated through indirect citation chains long after retraction. 7 The Wakefield paper — which falsely claimed a link between the MMR vaccine and autism — was retracted in 2010 but had already generated a citation network that continued to propagate its findings for years afterward, with documented public health consequences.

The mechanism is straightforward: Paper A is retracted. Papers B, C, and D, which cited Paper A before retraction, are not retracted. Papers E, F, and G cite B, C, and D without knowing that the foundational claim in A has been invalidated. The error propagates through the network indefinitely.

3.5 The Unknown Unknown Problem

The fifth and most philosophically significant failure mode is not a distortion of existing knowledge but an absence of knowledge that is not acknowledged as such. This is the unknown unknown — the question that has not been asked, the combination of variables that has not been studied, the gap in the evidence space that is invisible because nobody has mapped it.

In the citation system as it currently exists, there is no mechanism for citing an unknown unknown. A paper either cites a source or it does not. There is no standard way to say "this claim exists at the boundary of current evidence, and here is the precise shape of what we do not yet know." The result is that claims at the edge of knowledge are cited the same way as claims at the centre of it — with the same confidence, the same format, and the same apparent authority.

This is the problem that no existing citation tool addresses.

4. The Current Tool Landscape

The market for citation verification tools has developed in two distinct tiers, with a significant gap between them and the problem they collectively fail to solve.

Tool	Core capability	Failure mode addressed	Buyer	Domain depth
CiteTrue	Checks whether a citation exists	Fabrication	Students, academics	None

Citely	Checks existence across CrossRef, PubMed, arXiv	Fabrication	Students, academics	None
Recite	Checks formatting and existence	Fabrication	Students, academics	None
Scite.ai	Classifies citations as supporting, contrasting, or mentioning	Amplification (partially)	Researchers, pharma, institutions	None
Semantic Scholar	Citation graph analysis and paper discovery	None (discovery tool)	Researchers	None
Undermind	Deep literature search across citation graphs	None (discovery tool)	Researchers	None
Westlaw KeyCite	Checks whether a case is still good law	Retraction equivalent	Legal professionals	Legal
Harvey AI / Casetext	Legal research and drafting assistance	None (drafting tool)	Legal professionals	Legal

Scite.ai is the most sophisticated existing player. Founded in 2018 and acquired by Research Solutions in 2024, Scite has indexed 1.6 billion citation statements, established direct licensing agreements with 40+ publishers including Wiley, SAGE, and the BMJ, and serves over 2 million users including Novo Nordisk, Stanford, and ETH Zurich. Their Smart Citations product classifies each citation as supporting, contrasting, or mentioning, with the surrounding sentence shown in context. They have an MCP integration for Claude and ChatGPT. ⁸

What Scite does for a company like Novo Nordisk is primarily competitive intelligence and research monitoring: tracking how their drugs and compounds are being cited across the literature, identifying key opinion leaders, and receiving alerts when papers supporting their assets are retracted or receive editorial notices. This is valuable but it is a monitoring tool, not a claim integrity tool. It tells you how a paper is being cited; it does not tell you whether a specific commercial or regulatory claim is accurately supported by the paper it cites.

The gap that no tool in this table addresses is the one identified in Section 3: **whether a specific claim — in a marketing document, a regulatory submission, an LLM output, or a legal brief — accurately represents what its cited source actually found.**

5. The AI Dimension

The arrival of large language models has transformed the citation problem from a quality issue into a systemic integrity crisis. The mechanisms are distinct from traditional citation failure and require separate analysis.

5.1 Hallucination at Scale

The fabrication problem, which was previously limited to individual cases of fraud or error, is now operating at industrial scale. The 2026 arXiv study "LLM Hallucinations in the Wild" estimated 146,932 hallucinated citations in papers published in 2025 alone, with the social sciences showing the highest rates. ² The GhostCite study identified patterns of repeated invalid citations — the same fabricated paper appearing across multiple AI-generated documents — suggesting that LLMs are not generating random errors but are reproducing specific hallucinated citations from their training data. ⁹

5.2 Misrepresentation at Scale

More significant than fabrication is the misrepresentation problem. An LLM asked to summarise a paper will frequently produce a summary that is directionally correct but quantitatively wrong, omits key caveats, overstates the strength of the evidence, or applies findings from one context to another. When this summary is then cited in a paper or document, the citation points to a real paper — it passes every existence check — but the claim it supports is not what the paper actually demonstrated.

Research from MIT in early 2025 found that AI models use more confident language when hallucinating than when stating facts, meaning that the most confidently stated AI claims are statistically more likely to be wrong. ¹⁰ This creates a systematic bias toward overconfident misrepresentation that is invisible to any tool that only checks citation existence.

5.3 The RAG Partial Solution

Retrieval-Augmented Generation (RAG) is the current industry response to LLM citation problems. RAG systems retrieve relevant documents before generating a response, grounding the output in actual sources rather than training data. This substantially reduces fabrication. It does not solve misrepresentation — an LLM can retrieve the correct paper and still mischaracterise what it says. And it does not solve the unknown unknown problem

— an LLM cannot tell you that no evidence exists for a specific combination of variables, because it has no map of the evidence space.

The VeriCite system (2025) represents the current frontier of RAG-based citation verification, using a rigorous post-generation validation step to check whether generated citations actually support the claims they are attached to. ¹¹ This is a meaningful advance but it operates at the level of individual documents, not at the level of the knowledge graph.

6. The Graded Citation: A New Epistemic Primitive

The research above points toward a gap that is not merely commercial but conceptual. The citation system, as it has existed for four centuries, has only one epistemic status: cited or not cited. A claim either has a reference or it does not. This binary is inadequate for the complexity of the actual evidence landscape.

What is needed is a **graded citation** — a citation that carries information not just about what source is being referenced, but about the epistemic status of that reference in relation to the specific claim being made. A graded citation would have four possible statuses:

Verified — the cited source directly and accurately supports the claim as stated, with appropriate confidence and scope. The claim is a known known.

Contested — the cited source exists and is relevant, but the evidence is disputed, qualified, or inconsistent across the literature. The claim is a known known with contested confidence.

Implied but untested — no source directly addresses the claim, but the knowledge graph contains connected evidence that implies the claim may be true. The claim is an unknown known — knowledge that exists in distributed form but has not been synthesised or directly tested.

Beyond current evidence — no source addresses the claim, and the knowledge graph contains no implied connections. The claim is an unknown unknown. The honest citation here is not a reference to a paper but a structured acknowledgment of the boundary of knowledge.

This fourth status is the one that is entirely absent from the current citation system. There is no standard way to say "this claim has not been tested and here is what we do know about the adjacent territory." The result is that claims at the boundary of knowledge are either unsupported (no citation) or falsely supported (a citation that does not actually address the claim). Both are failures of epistemic honesty.

7. The Opportunity for truthdesk.claims

The architecture of truthdesk.claims maps directly onto the graded citation framework described above.

The platform's verdict engine already produces four verdict types — Supported, Contradicted, Partially Supported, and Insufficient Evidence — which correspond closely to the four epistemic statuses of the graded citation. The knowledge graph, built from continuous ingestion of PubMed, PMC, PDB, UniProt, OpenFDA, and ClinicalTrials.gov, is the infrastructure needed to determine not just whether a source exists, but what the topology of the evidence space looks like around any given claim.

The Frontier Engine — which generates new testable hypotheses by identifying gaps and implied connections in the knowledge graph — is the mechanism for surfacing unknown knowns: claims that are implied by the existing evidence but have not yet been directly tested. This is a capability that no existing citation tool possesses.

The Dream State Engine — which runs contradiction simulations and stress-tests the knowledge graph — is the closest existing mechanism for probing unknown unknowns: identifying the questions that the current evidence cannot answer and characterising the shape of that ignorance.

The competitive position is therefore not in the tier occupied by CiteTrue or Citely (fabrication detection) or even in the tier occupied by Scite (citation context classification). The competitive position is in a new tier: **claim integrity against a structured knowledge graph, with explicit representation of the boundary of knowledge.**

The buyers for this capability are distinct from the buyers for existing citation tools. They are not students checking their bibliographies. They are regulatory affairs teams at pharmaceutical companies, legal professionals reviewing brief accuracy, AI governance teams auditing LLM outputs, and regulators assessing the substantiation of health claims. Each of these buyers needs not just "does this citation exist" but "does this citation accurately support this specific claim, and what is the epistemic status of that support."

8. The Domains

The graded citation framework is domain-agnostic. The same underlying capability — fetch the source, find the relevant passage, compare it to the claim being made, return a structured verdict with epistemic status — applies across every domain where citations are used to assert truth.

Life sciences and pharma is the founding domain. The regulatory tailwind is strong: the FDA, EMA, and EFSA are all tightening substantiation requirements for health claims. The Novo Nordisk use case — monitoring how research assets are cited — is the current state of

the art. The next state of the art is verifying whether regulatory submissions accurately represent the evidence they cite.

Legal is the largest adjacent domain. The Harvard Caselaw Access Project and CourtListener provide free full-text access to millions of cases. The specific problem — whether a brief accurately represents what a case held — is documented, consequential, and unaddressed by any existing tool. Courts are already issuing standing orders requiring AI certification. The professional responsibility dimension is growing.

AI governance is the fastest-growing domain. Every enterprise deploying LLMs for research, compliance, or decision support needs to audit whether LLM outputs accurately represent their cited sources. This is becoming a regulatory requirement under the EU AI Act and is increasingly a board-level risk management question. The market is large and the tooling is nascent.

Academic publishing is the domain with the most immediate crisis. 146,932 hallucinated citations in 2025 alone. Major publishers requiring AI disclosure. Journal editors looking for automated integrity checks. The problem is acute and the existing tools address only fabrication, not misrepresentation.

9. Conclusion

The citation has been the fundamental unit of scholarly communication for four centuries. It began as a courtesy, became an obligation, and was transformed by Garfield into a currency. It is now under pressure from three directions simultaneously: the industrialisation of citation distortion through LLMs, the growing complexity of the evidence landscape that binary citation cannot represent, and the regulatory and legal demands for higher standards of substantiation.

The response to this pressure requires more than better existence-checking. It requires a new epistemic primitive — the graded citation — that carries information about the quality, scope, and boundary of the evidence it represents. The unknown unknown is not a failure of citation; it is the honest acknowledgment that knowledge has a frontier, and that frontier deserves to be cited as clearly as the settled territory behind it.

That is what truthdesk.claims is building.

References

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